

1. Administrative & Study Identification

Full Study Title: Real-World Comparative Effectiveness Research and Related Economic Outcomes Among Nonvalvular Atrial Fibrillation Patients Using Oral Anti-coagulants
Short Title/Acronym: ARISTOPHANES (Anticoagulants for Reduction In Stroke: Observational Pooled Analysis on Health Outcomes and Experience of Patients)
Protocol Number: CV185-543
NCT Number: NCT03087487
Sponsor Name: Bristol-Myers Squibb
Investigational Product: Oral Anticoagulants / OACs (Apixaban, Dabigatran, Rivaroxaban, Warfarin)
Phase of Development: N/A (Observational / Retrospective Database Analysis)
Medical Monitor: N/A (Retrospective Claims Data Study)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To compare the risk of major bleeding (MB) and stroke/systemic embolism (SE) events among oral anticoagulant (OAC)-naïve non-valvular atrial fibrillation (NVAF) patients initiating OAC treatment with warfarin, apixaban, dabigatran, or rivaroxaban. **Secondary Objectives:** To compare the effectiveness and safety of non-vitamin K antagonist oral anticoagulants (NOACs) versus each other and versus warfarin in specific subpopulations (e.g., patients aged $\geq$ 80 years, obese patients, and those with active cancer or diabetes mellitus).

2.2 Background & Rationale To provide a clearer understanding of the effectiveness and safety associated with various oral anticoagulant treatment options in NVAF populations within routine clinical practice. Randomized clinical trials may not fully represent real-world clinical experience, particularly for high-risk elderly patients and those with concurrent morbidities like active cancer or extreme obesity. This large-scale, real-world data analysis uses pooled CMS Medicare and commercial claims databases to inform standard-of-care practices.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Retrospective cohort study / Database analysis) **Control Type:** Active Comparator (Propensity Score Matched comparisons among Apixaban, Dabigatran, Rivaroxaban, and Warfarin) **Blinding:** Open-label / Non-blinded

(Claims Data) **Randomization:** N/A (1:1 NOAC-Warfarin and NOAC-NOAC propensity score matching used for analysis)

3.2 Planned Enrollment & Duration Target **SN\$:** 466,991 total participants (pooled from US Medicare and 4 commercial claims databases) **Number of Sites:** N/A (Retrospective Database Study utilizing MarketScan, PharMetrics, Optum, and Humana databases, along with CMS Medicare) **Estimated Study Start:** 01-Jun-2016 **Estimated Primary Completion:** 08-Mar-2023

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Patients aged ≥ 18 years old as of the index date. 2. At least 1 diagnosis of atrial fibrillation prior to or on the index date, identified by any medical claim. 3. Had ≥ 1 pharmacy claim for apixaban, dabigatran, rivaroxaban, or warfarin during the identification period (01-Jan-2013 to 30-Sep-2015). The first OAC pharmacy claim date during this period is designated as the index date. 4. At least 12 months of baseline period prior to the index date with continuous medical and pharmacy enrollment.

4.2 Exclusion Criteria 1. Evidence of valvular heart disease, transient atrial fibrillation (pericarditis, hyperthyroidism, thyrotoxicity), venous thromboembolism, or heart valve replacement/transplant during the 12-month baseline period or on the index date. 2. Evidence of pregnancy during the study period. 3. Had a pharmacy claim for warfarin, apixaban, dabigatran, rivaroxaban, or edoxaban during the 12-month baseline period (i.e., non-naive OAC users). 4. Had > 1 oral anticoagulant claim on the index date. 5. Hip or knee replacement surgery within 6 weeks before the index date.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Standard or reduced dosing of prescribed NOAC (Apixaban, Dabigatran, Rivaroxaban) or Warfarin as determined by treating physicians in real-world practice. **Route of Administration:** Oral. **Duration of Treatment:** Variable. Patients were followed from the day after the index date until the earliest of: 30 days after discontinuation, medication switch, death, end of continuous enrollment, or end of the study period (30-Sep-2015).

5.2 Concomitant & Prohibited Therapy Permitted: Routine concomitant therapies prescribed in real-world clinical practice for NVAf and comorbidities. **Prohibited:** Concomitant use of multiple initial OAC therapies (excluded on index date).

6. Study Timeline & Visit Schedule

(Note: As a retrospective claims database study, traditional onsite visits do not apply. Timeline represents data extraction windows.)

Visit ID	Window	Key Activities/Procedures
Baseline Period	12 Months Prior to Index	Verification of continuous enrollment; capture of comorbidities; confirmation of OAC-naïve status and exclusion criteria.
Index Date	Day 0 (01-Jan-2013 to 30-Sep-2015)	First prescription claim of apixaban, dabigatran, rivaroxaban, or warfarin. Initiation of follow-up.
Follow-up	Day 1 to End of Follow-up	Tracking of hospitalization claims for efficacy (Stroke/SE) and safety (Major Bleeding) endpoints until treatment discontinuation, switch, death, or end of study (30-Sep-2015).

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Risk of stroke/systemic embolism (SE) events and major bleeding (MB). **Measurement Method:** Identified retrospectively via hospitalizations with stroke/SE or MB as the principal or first-listed diagnosis using ICD-9-CM/ICD-10-CM codes in medical claims data.

7.2 Secondary & Exploratory Endpoints Rates of composite thrombotic events (stroke/SE, myocardial infarction, and death). Comparative risks of Stroke/SE and MB across subgroups stratified by extreme age (≥ 80 years), active cancer status, diabetes mellitus, and extreme body weight/obesity classes.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Extracted retrospectively from administrative health claims data. **Serious Adverse Events (SAEs):** Captured through records of inpatient hospital admissions for major bleeding, hemorrhagic stroke, or other fatal/life-threatening events. **Data Monitoring Committee (DMC):** N/A (Observational, de-identified retrospective database study).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Eligible NVAf patients who initiated targeted OACs during the identification window, analyzed using 1:1 Propensity Score Matching (PSM) for NOAC-warfarin and NOAC-NOAC comparisons. **Sample Size Justification:** Descriptive and comparative real-world analyses utilizing all eligible NVAf subjects ($N=466,991$) across 5 databases (CMS Medicare + 4 commercial claims databases) without prospective power calculations. **Handling of Missing Data:** Required at least 12

months of continuous medical and pharmacy enrollment prior to the index date to assure data completeness; Cox proportional-hazards models used for matched cohorts.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: The study analysis was conducted using de-identified patient records in compliance with the Health Insurance Portability and Accountability Act (HIPAA) regulations for observational research. **Monitoring Plan:** N/A for retrospective claims analysis. **Audit Trail:** Validated leveraging of CMS Medicare and IMS PharMetrics, Optum, Humana, and Truven MarketScan databases with transparent ICD coding definitions.

11. Study Identifiers & Governance

Primary ID: CV185-543 **Registry Number:** NCT03087487 **Sponsor/Lead Organization:** Bristol-Myers Squibb **Study Title:** Clinical and Economic Outcomes of Oral Anticoagulants in Non-valvular Atrial Fibrillation **Official Regulatory Title:** Real-World Comparative Effectiveness Research and Related Economic Outcomes Among Nonvalvular Atrial Fibrillation Patients Using Oral Anti-coagulants

12. Clinical Framework (Atrial Fibrillation Specific)

Primary Condition: Non-valvular Atrial Fibrillation (NVAF) **Diagnosis Threshold:** ≥ 1 medical claim for Atrial Fibrillation prior to or on the index date without evidence of valvular heart disease **Medical Methodology:** Initiation of Oral Anticoagulant Therapy (DOACs or VKAs) **Optimization Target:** Prevention of stroke/systemic embolism while minimizing major bleeding risks

13. Study Procedures & Methodology

Management Pathway: Standard routine clinical practice for NVAF (claims-derived) **Education Protocol (HCP):** N/A (Observational Study) **Education Protocol (Patient):** N/A (Observational Study) **Quality Audit Mechanism:** Rigorous Propensity Score Matching (PSM) adjusting for baseline demographics, clinical comorbidities, and medication history to balance patient cohorts

14. Observation & Follow-Up Schedule

Total Duration: Variable per patient (tracked from index date up to 30-Sep-2015 limit) **Onsite Visit Cadence:** N/A (Retrospective

Database) **Remote Monitoring Frequency:** Continuous evaluation of administrative claims **Checklist Review Interval:** N/A

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				